

Matvei Borodin

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Objective

ML Engineer focused on deep learning, LLMs, NLP, and large-scale ML systems. Experienced in production ML for generative search and voice interaction systems, with interests in RL, alignment, RAG, and reliable reasoning in LLMs. Strong foundation in algorithms and applied ML; proficient in Python, C++, and PyTorch.

Experience

Alice Voice Activation Team

Moscow, Russia

ML Engineer Intern, Yandex

May 2026 — Present

- Contributed to **speech/SLU models** for Alice voice activation, including **data preparation**, **model training**, inference-related code, and quality analysis.
- Improved **quick commands** for music and smart home control by experimenting with data and model architectures to increase recognition quality for production **voice interaction systems**.

Alignment Team for Alice Search Generative Answers

Moscow, Russia

ML Engineer Intern, Yandex

December 2025 — March 2026

- Worked across the **end-to-end pipeline** of generative answers for search and contributed in two directions: **development** of alignment infrastructure and **research** on RL training artifacts for the **RAG agent**.
- Developed components of the alignment and metrics infrastructure: integrated a new **factchecking reward**, implemented a **GPU whitelist** for quantized reward models, fixed versioning/build issues, corrected aggregation of retrieved information contexts produced by the **search planner**, and contributed to runtime infrastructure for **online GRPO** reward computation.
- Conducted research-oriented experiments with **CERL**, **DPO**, and **GRPO**: studied how training data composition and hyperparameters affected model behavior, analyzed **GRPO artifacts** using **side-by-side** assessor evaluations, and identified **reward hacking** patterns and quality regressions.

A Russian Multi-hop QA Benchmark for LLM Search and Reasoning

Moscow, Russia

Pretrain YandexGPT Analytics, YSDA (School of Data Analysis)

October 2025 — Present

- Developed a Russian-language **multi-hop QA benchmark** focused on complex factual queries where LLMs must combine several constraints, reason over relations, and use search-based retrieval instead of relying only on parametric memory.
- Constructed benchmark slices targeting key LLM capabilities and failure modes: **Reversal Curse**, constraint preservation, list-style answering, hallucinations in no-answer cases, and retrieval on rare facts.
- Evaluated standalone LLMs and search-based agentic systems, including **YandexGPT Pro 5.1**, baseline models, **Alice Search**, and **NeuroAlice**, separating parametric memory, reasoning, and retrieval-based behavior.

Education

Higher School of Economics

Moscow, Russia

Bachelor's Degree, Faculty of Computer Science

2022 — 2026

Academic Achievements

Training: Reinforcement Learning ML Run | Yandex

September 2025

Studcamp: Mathematics in AI, team optimization project | Yandex, YSDA, SPbU

November 2025

A/B Week, Yandex School of Data Analysis (YSDA) | Yandex

July 2025

Winner, All-Siberian Open School Olympiad in Mathematics

2020 — 2021

Technical Skills

Languages: Python, C++, Swift, Kotlin, Java

Technologies: PyTorch, SQL, Nirvana (yandex apache airflow analog), CatBoost, LoRA, Git

Skills: Machine Learning, Deep Learning, Classical ML, NLP, LLMs, RL (CERL, PPO, DPO, GRPO), RAG